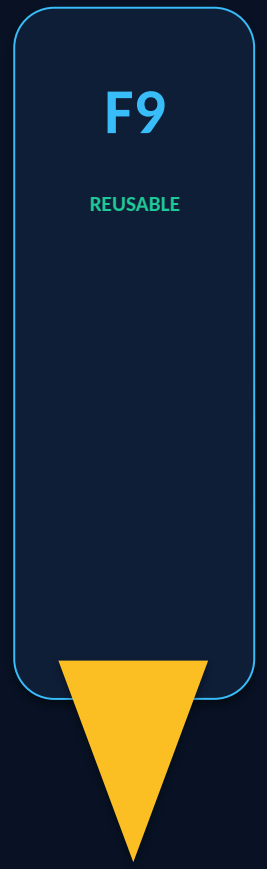
DATA SCIENCE CAPSTONE

Predicting Falcon 9 First-Stage Landing Success

An end-to-end data science journey from API collection to predictive modeling



Priti Umbare | IBM Applied Data Science Capstone

Executive Summary

LAUNCHES	FEATURES	TEST SET	DT ACCURACY
90	83	18	83.33%
Falcon 9 records after filtering	After one-hot encoding	20% held-out sample	Course test result

- Objective: predict whether the Falcon 9 first stage will land successfully.
- Collected launch data through the SpaceX API and web scraping, then cleaned and transformed it for analysis.
- EDA and SQL revealed relationships between launch experience, launch site, orbit and landing outcomes.
- Interactive Folium and Plotly Dash tools turned the analysis into decision-friendly exploration.
- Classification models were tuned and evaluated; the capstone validation identifies the linear SVM kernel as the best SVM kernel.

Source: IBM Skills Network capstone workflow; SpaceX public launch data

Introduction: why landing success matters

THE BUSINESS QUESTION

Can historical launch data predict the probability that a Falcon 9 first stage will land successfully?

- Successful recovery enables booster reuse and supports lower launch economics.
- Understanding the drivers of success helps compare launch configurations and assess risk.
- The project turns a real aerospace problem into a reproducible data science workflow.



Outcome: a complete analytical narrative

Collect → clean → explore → visualize → map → dashboard → model → evaluate

End-to-end methodology

1

Collect

SpaceX API + web scraping

2

Wrangle

Missing values + labels + encoding

3

Explore

EDA + SQL + visual analytics

4

Interact

Folium map + Plotly Dash

5

Predict

LR + SVM + DT + KNN

Design principle: every method answers a specific question and feeds the next stage.

Data collection & data wrangling

SOURCE 01

SpaceX REST API

Launch metadata, booster details, payloads, outcomes

SOURCE 02

Wikipedia scraping

Historical launch records and complementary fields

TARGET

Class = 1 / 0

1 = successful landing; 0 = unsuccessful

WRANGLING PIPELINE

- Filter the dataset to Falcon 9 launches.
- Identify missing values and impute PayloadMass where required.
- Convert mission outcomes into a binary landing-success label.
- Select numerical and categorical predictors relevant to landing performance.
- Apply one-hot encoding to Orbit, LaunchSite, LandingPad and Serial.
- Standardize features before classification modeling.

Final modeling matrix: 83 features; 80/20 split → 72 training + 18 test records.

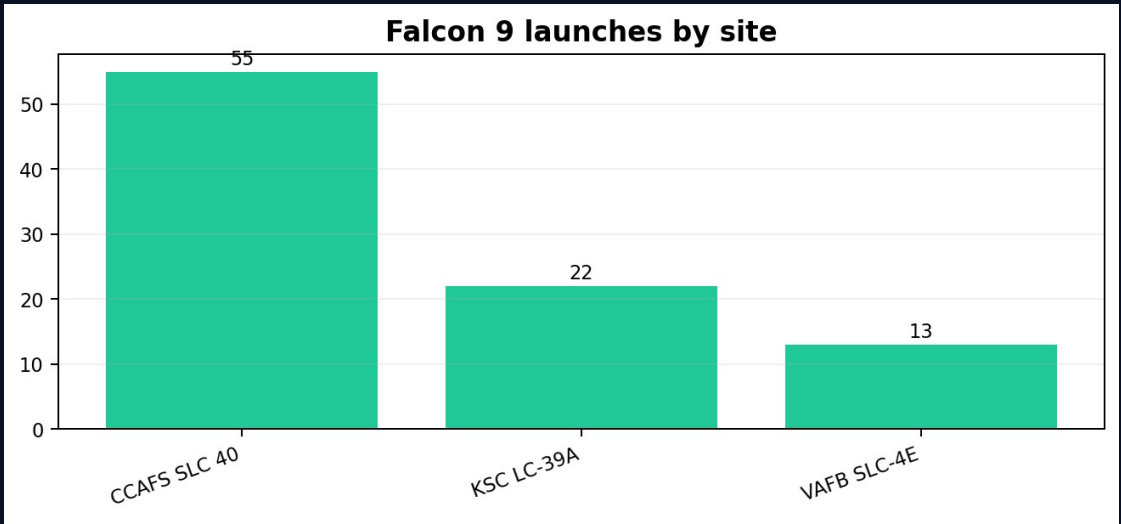
EDA & interactive visual analytics methodology

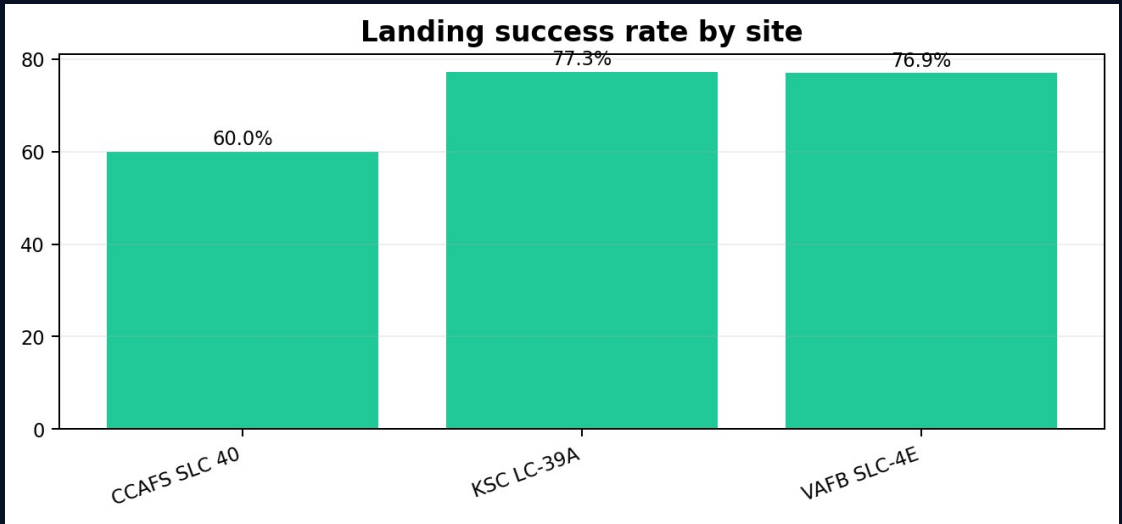
- Univariate counts: launch sites, orbit types and mission outcomes.
- Relationship analysis: FlightNumber, PayloadMass, LaunchSite and Orbit.
- Trend analysis: landing success over time.
- SQL: aggregation, filtering, minimum/maximum and date-based questions.
- Folium: geographic distribution, markers and clustering.
- Plotly Dash: site selection, payload range filtering and interactive outcome views.

Question	Method	Output
Where do launches occur?	GroupBy / SQL	Site counts
Which configurations succeed?	EDA plots	Success-rate patterns
Where are sites located?	Folium	Interactive map
How does payload affect outcome?	Dash slider	Filtered scatter
Can success be predicted?	Classification	Test accuracy

Source: IBM Skills Network capstone workflow; SpaceX public launch data

EDA with visualization — results





CCAFS SLC 40 carried the most launches; KSC LC-39A had the highest success rate among the three main sites.

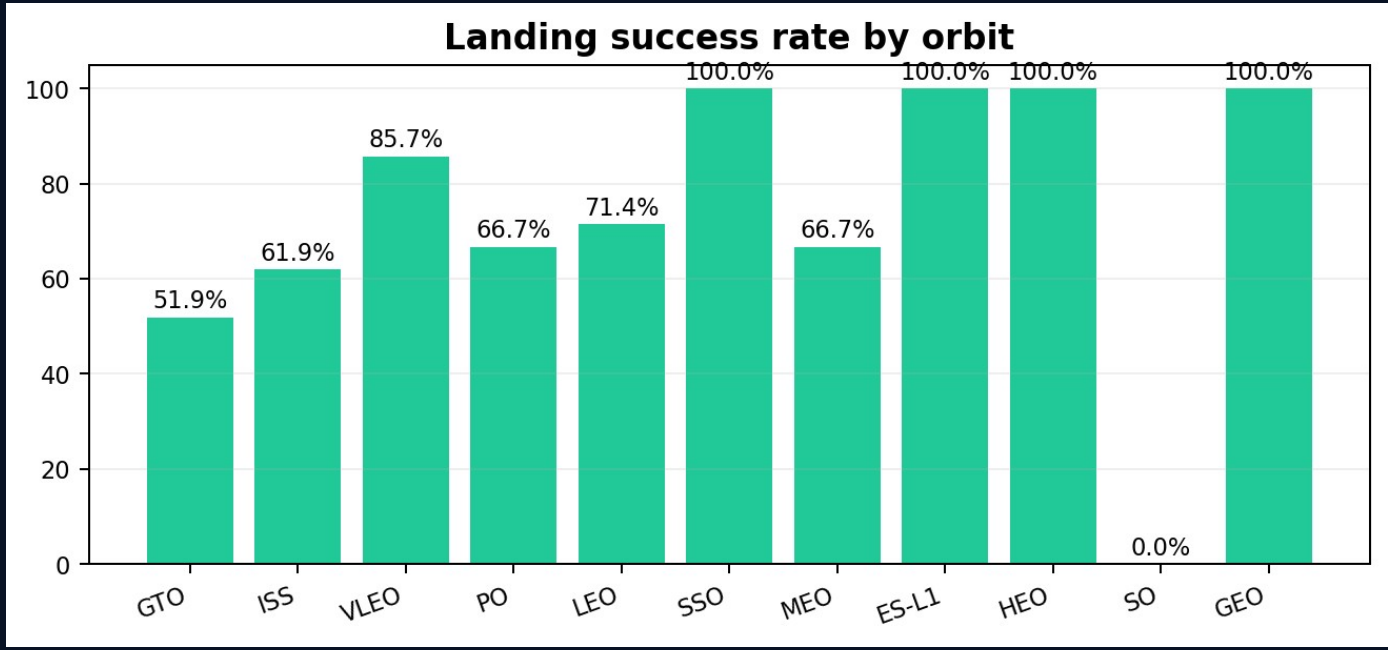
The strongest visual pattern is temporal: early flight numbers contain more failures, while later flights show substantially more successful landings.

EDA with SQL — results

SQL question	Result
Unique launch sites	CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E
Total NASA (CRS) payload	45,596 kg
Average F9 v1.1 payload	2,928.4 kg
First successful ground-pad landing	2015-12-22

SQL converted raw records into compact, auditable answers that complement the visual EDA.

Orbit-level results



- GTO was the most common orbit (27 missions).
- Several low-count orbit categories show 100% observed success; these should be interpreted cautiously because their sample sizes are tiny.
- VLEO shows a strong success rate (85.71%) in the analyzed sample.
- Orbit is informative, but it should be interpreted together with flight experience and launch configuration.

Key lesson: rate + sample size must be read together.

Interactive map — Folium results

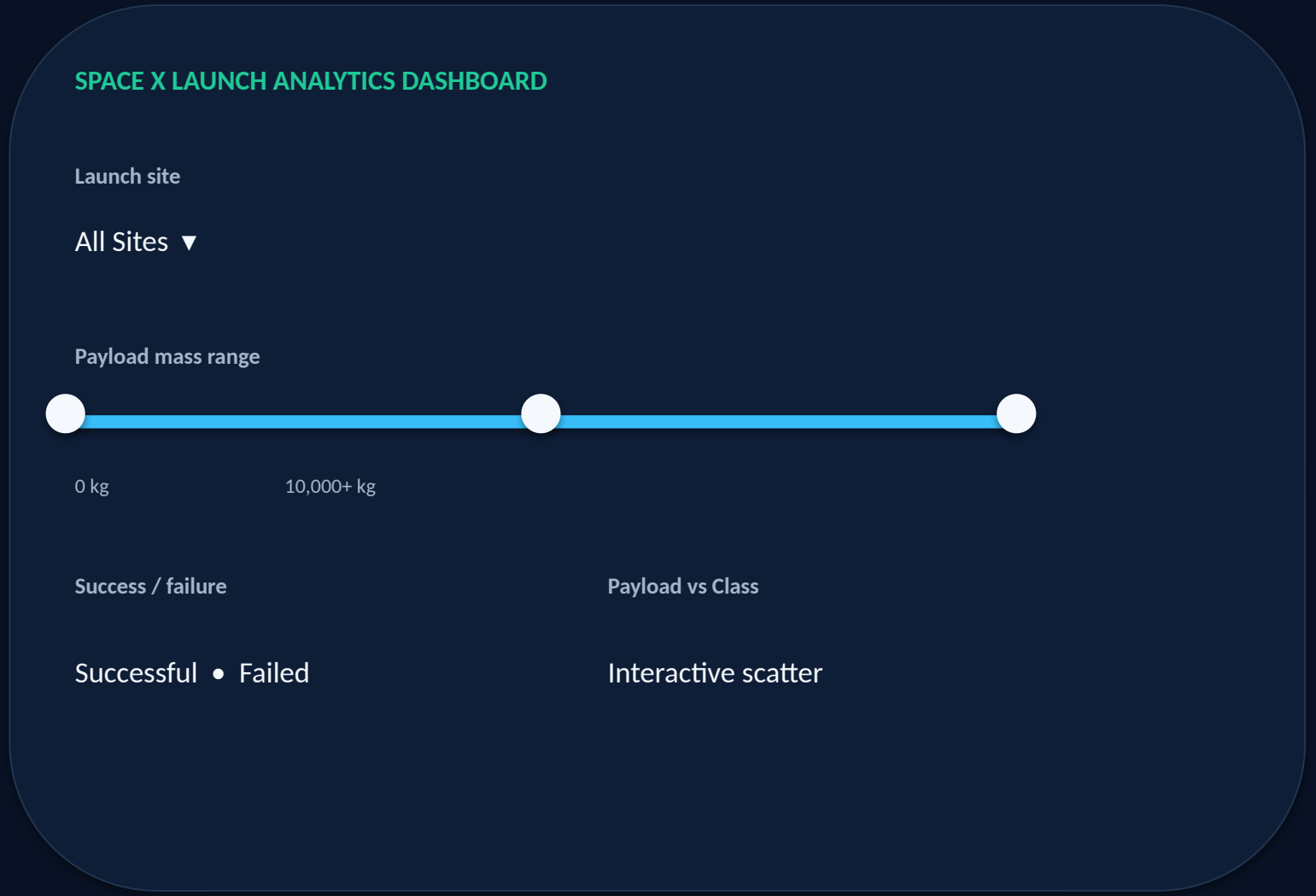
FOLIUM LAUNCH-SITE VIEW



- Markers identify launch locations.
- MarkerCluster reduces overlap when many launches share nearby coordinates.
- Success/failure can be encoded visually for rapid spatial comparison.
- Geographic context helps explain why launch facilities are concentrated near coastlines.

Interactive takeaway: location becomes a filterable analytical dimension—not just a map.

Plotly Dash dashboard — methodology



- Dropdown: choose a launch site.
- Range slider: restrict payload mass.
- Pie chart: compare landing outcomes.
- Scatter plot: inspect payload vs. success by booster/version.

The dashboard converts static analysis into a decision interface.

Plotly Dash — results

BEST SITE

KSC LC-39A

Highest observed success rate among the main sites

SUCCESS RATE

76.9%

22 launches; 17 successful in the analyzed table

PAYLOAD RANGE

~1.95–5.3 t

Highest observed success band in dashboard analysis

- Site-level filtering makes differences in landing outcomes immediately visible.
- Payload filtering helps isolate ranges where successful recovery is more common.
- Booster-version segmentation reveals the importance of reusable hardware generations.
- The dashboard is most useful as an exploratory tool; it does not replace the statistical model.

Source: capstone dashboard analysis; figures summarized from the project workflow

Predictive analysis — methodology

Logistic Regression	SVM	Decision Tree	KNN
Linear baseline	Kernel comparison	Non-linear rules	Distance-based

- 80/20 train-test split with random_state = 2.
- StandardScaler applied to the feature matrix.
- GridSearchCV used to tune model hyperparameters.
- Evaluation based on validation performance and held-out test accuracy.

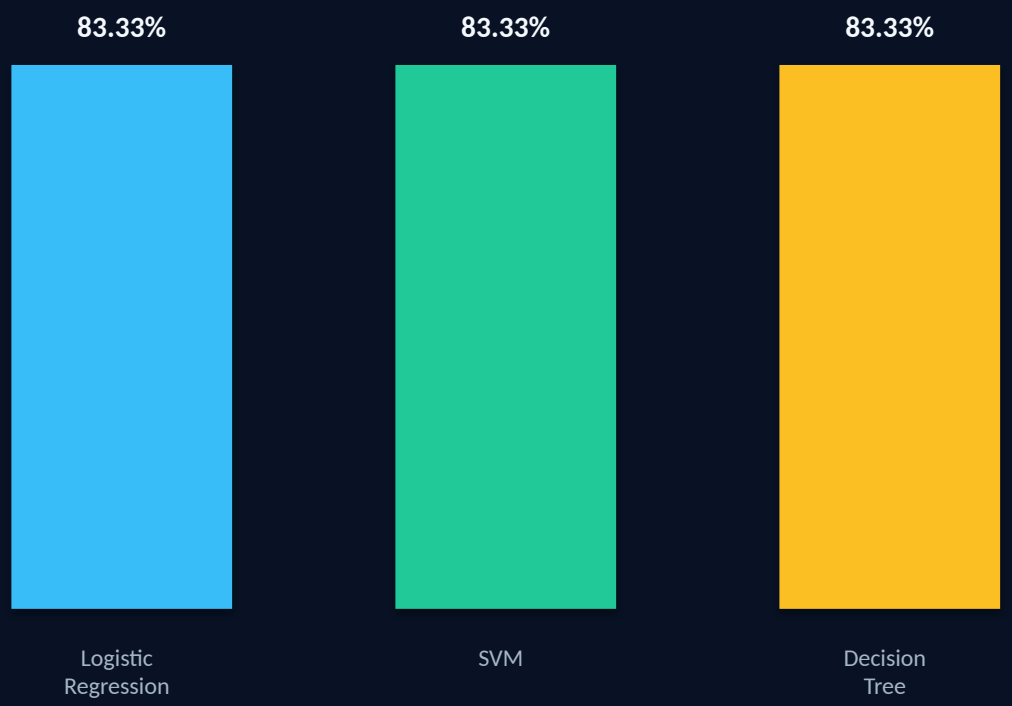
SVM VALIDATION RESULT

Best kernel: LINEAR

Test sample = 18 records
Small test size means accuracy should be interpreted with care.

Predictive analysis — classification results

TEST ACCURACY (COURSE RUN)



INTERPRETATION

- Decision Tree test accuracy: 83.33%.
- SVM test accuracy: 83.33%; linear kernel selected by the capstone validation quiz.
- With only 18 test records, one prediction changes accuracy by about 5.56 percentage points.
- Model selection should consider validation behavior and interpretability—not just one test score.

Recommended reporting practice: pair accuracy with the confusion matrix and validation score.

Innovative insights

01	**Experience effect**	Later FlightNumbers show substantially more successful landings than early missions.
02	**Site effect**	KSC LC-39A leads the main sites in observed success rate.
03	**Orbit effect**	Orbit type changes the operating context; low-count 100% rates need cautious interpretation.
04	**Payload effect**	A middle payload band (~1.95–5.3 t) shows the strongest observed success in the dashboard analysis.
05	**Reusability signal**	Booster reuse, grid fins, legs and later blocks are strong candidates for predictive features.

Conclusion

A complete data science pipeline can turn launch history into operational insight.

- Data collection combined API and web sources into a usable Falcon 9 dataset.
- EDA + SQL explained where, when and under what configurations landings succeed.
- Folium and Dash made geographic and configuration-level exploration interactive.
- Classification demonstrated that landing success can be predicted with useful accuracy, while highlighting uncertainty from a small test set.
- The most valuable insight is not a single score: it is the repeatable workflow from raw launch data to evidence-based decision support.

Next step: expand the dataset with newer launches and evaluate calibration, recall and cost-sensitive metrics.

Repository & submission checklist

GITHUB REPOSITORY

https://github.com/priti-insights/falcon9-landing-prediction

Required repository contents

- Data collection / web scraping notebooks
- Data wrangling + EDA + SQL notebooks
- Folium map notebook
- Plotly Dash application
- Machine learning notebook
- README with project overview
- Dataset / generated outputs as permitted
- Charts or screenshots
- Reproducible execution order
- Final PDF report

Submission filename: Data Science Capstone Project Report.pdf

Appendix — key numbers

Metric	Value
Falcon 9 records	90
One-hot encoded features	83
Training / test	72 / 18
GTO missions	27
CCAFS SLC 40 launches	55
KSC LC-39A launches	22
VAFB SLC-4E launches	13
NASA (CRS) total payload	45,596 kg
Average F9 v1.1 payload	2,928.4 kg
First successful ground-pad landing	2015-12-22
Decision Tree test accuracy	83.33%
SVM best validation kernel	linear

Interpretation note

The capstone uses a small historical dataset. Reported percentages are sensitive to individual test observations, so model validation and domain context matter alongside raw accuracy.

End of report
Thank you